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21 June 1978
SUPERSEDING
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3 February 1978

MILITARY SPECIFICATION

CLOTH, NONWOVEN: INTERLINING, FUSIBLE, NYLON AND POLYESTER, POLYAMID

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

*1.1 Scope. This specification covers nonwoven fusible cloth.

*1.2 Classification. The fusible cloth shall be one type in the following classes as specified (see 6.2):

Class 1 - White
Class 2 - Tan
Class 3 - Oxford Grey

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commandant of the Marine Corps, Headquarters U. S. Marine Corps (Code LMA-1), Washington, D.C. 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal, shall apply.

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified (see 6.2), the contractor shall furnish sample cloth for first article inspection and approval. The first article shall consist of 5 yards, full width. The approval of the first article authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all provisions of the specification. The first article shall be manufactured by the contractor in the same facilities to be used for the manufacture of the production cloth.

3.3 Material.

3.3.1 Fibers. The fiber shall be polyethylene terephthalate and nylon. The finished cloth shall contain 50 ±5 percent polyester and the remaining percentage nylon, when tested as specified in 4.6.

3.3.2 Web. The cross-laid webs shall be a blend of 50 percent polyester and 50 percent nylon (see 3.3.1). The minimum tensile strength, based on a 2-inch strip, in the machine direction and in the crosswise direction shall be as specified in Table I.

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3.4 Design and construction.

*3.4.1 Physical requirements. The finished cloth shall conform to the requirements shown in Table I when tested as specified in 4.6.

Table I. Physical requirements

Characteristic		Requirement
Weight per sq. yd.		
Minimum (ounces)		1.7
Maximum (ounces)		2.2
Breaking/Tensile Strength		
Minimum		
Machine Direction	(pounds)	3.0
Crosswise Direction	(pounds)	2.0
Shrinkage		
Maximum		
After Fusing	(percent)	1%
After 20 Launderings	(percent)	1%
After 20 Dry Cleanings	(percent)	1%
Bond Strength (per inch of width)		
Minimum		
Initial	(ounces)	22/Splits 1/
After 20 Launderings	(ounces)	18/Splits 1/
After 20 Dry Cleanings	(ounces)	18/Splits 1/

1/ Splits means that during Test Method 5951 of FED-STD-191, the fusible material cannot be removed from the outershell, causing the fusible material to split from itself rather than separate from the outershell.

3.5 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 200 yards. The cloth shall be put up on full width rolls as specified in PPP-P-1134.

*3.6 Width. The minimum width shall be 35 inches, unless otherwise specified (see 6.2).

3.7 Color. The color shall be as specified (see 6.2) and shall match the standard sample (see 6.3).

*3.7.1 Matching. The color of the finished cloth shall match the standard sample under artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

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*3.7.2 Colorfastness. The finished cloth shall show fastness to laundering (after 3 cycles), perspiration, crocking, wet-dry cleaning, dry heat and light equal to or better than the standard sample. When no standard sample is available, the dyed and finished cloth shall show "good" fastness to laundering (after 3 cycles), perspiration, crocking, wet-dry cleaning, dry heat and light after 40 standard fading hours and shall show a Munsell Value for crocking not lower than 8.5 dry and 7.5 wet when tested as specified in 4.6.

3.8 Finish. The fusible fabric shall have a uniformly distributed dot system of a nylon polyamid type fusible adhesive paste applied to one side in accordance with good commercial practice (see 4.6).

3.8.1 Weight by percent. The total weight by percent of the finished cloth (fiber plus adhesive) shall conform to the following requirements:

Polyester fiber	36.15 percent
Nylon fiber	36.15 percent
Nylon adhesive	23.26 percent
Other matter	4.44 percent

3.9 Fiber identification. Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

3.10 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When first article is required, it shall be tested as specified in 4.6.

4.5 Quality conformance inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.5.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.5.2 Examination of the end item. Examination of the end item shall be in accordance with 4.5.2.1 through 4.5.2.4.

*4.5.2.1 Yard-by-yard examination. The required yardage of each roll shall be examined, and defects clearly noticeable at normal inspection distance (3 feet) shall be classified as listed. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the most serious defect shall be counted. A continuous defect shall be counted as one defect for each yard or fraction thereof in which it occurs. The sample unit shall be 1 linear yard. The acceptable quality level shall be 4.0 defects per 100 units for all defects listed. The sample size shall be in accordance with inspection level II of MIL-STD-105. The lot size shall be expressed in units of 1 yard each. The number of rolls selected shall be in accordance with Table II. An approximately equal number of yards shall be examined from each roll in the sample.

Defects

Hole, cut, tear or break
Abrasion or rub mark resulting in damage to fiber 1/
Crease or wrinkle, hard embedded 1/
Lump 1/
Spot or stains 1/
Embedded foreign matter 1/

1/ Clearly visible at normal inspection distance (approximately 3 feet).

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4.5.2.2 Overall examination. Each defect shall be counted no more than once in each roll examined. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number shall be as shown in Table II.

Defects

Width less than specified
Overall uncleanness
Objectionable odor
Not labeled in accordance with the Textile Fiber Products Identification Act

Table II. Sample size

Lot size (yards)	Sample size (rolls)	Acceptance number
Up to 1200 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

4.5.2.3 Examination for shade and finish. Each roll in the lot shall be examined visually for shade and finish (see 3.7 and 3.8). A roll shall be unacceptable if it fails to meet the requirements for shade match. The sample unit shall be a 4 inch by 20 inch swatch of the cloth. The sample unit shall be drawn from each roll in the lot.

4.5.2.4 Examination for length.

4.5.2.4.1 Individual rolls. The rolls shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number shall be as shown in Table II.

4.5.2.4.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the roll tickets. The rolls examined shall be those selected for the examination of individual rolls.

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4.5.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that the packaging, packing and marking complies with Section 5 requirements.

*4.6 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table III shall be followed. The physical and chemical values specified in Section 3, except when otherwise indicated, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be 5 yards full width of the finished cloth, and the lot size shall be expressed in units of 1 yard. The sample size shall be as follows and the lot shall be unacceptable if one or more sample units fail to meet any requirements specified:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table III. Text Methods

Characteristic	Requirement paragraph	Test Method
Fiber-Content		
Polyester (percent)	3.3.1	1/
Nylon (percent)	3.3.1	2530 1/
Color	3.7	Visual
Colorfastness to		
Laundering (after 3 cycles)	3.7.2	5614
Perspiration	3.7.2	5680
Crocking	3.7.2	5651
Wet-dry cleaning	3.7.2	5622
Light	3.7.2	5660
Dry Heat	3.7.2	5642
Weight	3.4.1	5041
Adhesive	3.8	1/
Breaking/Tensile Strength		
Machine Direction	3.4.1	5100
Crosswise Direction	3.4.1	5100

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Table III. Test Methods (cont'd)

Characteristics	Requirement paragraph	Test Method
Shrinkage		
After Fusing	3.4.1	Visual 2/
After 20 launderings	3.4.1	5552 2/
After 20 dry cleanings	3.4.1	5580 2/
Bond Strength		
Initial	3.4.1	5951 3/
After 20 launderings	3.4.1	5552/5951
After 20 dry cleaning	3.4.1	5580/5951 4/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ The fusible cloth shall be fused to the outershell (basic fabric) in accordance with 3/ prior to testing, then laundered in a wash and wear cycle 20 times and dried for 30 minutes with exhausted temperature of 140°F.

3/ The following fusing conditions of fusible material:

(a) Temperature between pressing surfaces of 300°F. minimum and 320°F maximum, at the fuseline (not the temperature at the clock).

(b) Pressure between pressing surfaces of 6 lb_f/in^2 minimum and 7 lb_f/in^2 maximum.

(c) Pressing time of 12 seconds minimum and 14 seconds maximum.

(d) Heat-dry.

4/ Pressing and shrinkage requirement shall not apply.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

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5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth is intended for use as fusible interlining material for clothing items.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number and date of this specification.
(b) Class (color) of cloth required (see 1.1).
(c) When first article is required (see 3.2).
(d) Length required if other than specified (see 3.5).
(e) Width of cloth required when other than specified (see 3.6).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

*6.2.2 Contract data requirements. Data conforming to Data Item Descriptions DI-R-4803, DI-R-4805, DI-T-4901, DI-T-4902, DI-T-4903 and DI-T-4904 will usually be required for delivery in connection with this specification for Marine Corps procurements. For other than Marine Corps, the cognizant procuring activity will provide comparable Data Item Descriptions. When so required, such data will be specified for delivery on a Form DD-1423 included in the contract.

6.3 Standard sample. For access to standard sample (see 3.1), address the procuring office issuing the invitation for bids.

6.4 First article. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

6.6. Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from previous issue were made. This was done as a convenience only and the Government assumes no resbility whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this

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document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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